

Simple ON/OFF TIMER

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Imagine leaving home in the evening and returning to find freshly cooked rice ready in the rice cooker. Or consider needing a power outlet socket for mobile phone charger that automatically turns off after a set time to prevent overcharging and extend lifespan of the mobile phone and charger. These scenarios can be made possible with a simple on/off timer that can be easily assembled at home.

This simple on-off timer can control any household electrical device that needs to be switched on or off after a specific duration. It employs inexpensive components wired in an analogue circuit, providing extended timer duration without the complexity of digital ICs or microprocessor programming. Higher loads can be controlled by selecting higher current relays, switches, and wires, or external contactors. EFY lab tested

the prototype on a breadboard, as shown in Fig. 1.

Circuit and working

Fig. 2 shows the circuit diagram of the simple on/off timer. It is built around transformer X1, bridge rectifier BR1, 12V voltage regulator LM7812 (IC1), dual op-amp LM358 (IC2), transistor BC547 (T1), 12V SPDT relay, one 1N4007 diode (D1), signal diode 1N4148 (D2), two LEDs (LED1, LED2), and a few other components.

The circuit has op-amp IC LM358 as the core component. A capacitor is charged at constant current to achieve a linear voltage rise in the capacitor, ensuring a consistent time interval. The 230V mains AC power is stepped down to 12V AC by transformer X1, rectified by BR1, and stabilised to 12V DC using filter capacitors C1 and C2, and the 12V voltage regulator LM7812.

Op-amp IC2A of the IC LM358 is configured as a unity gain buffer amplifier or voltage follower circuit. Here, the voltage at its output pin 1 follows the voltage at pin 3,

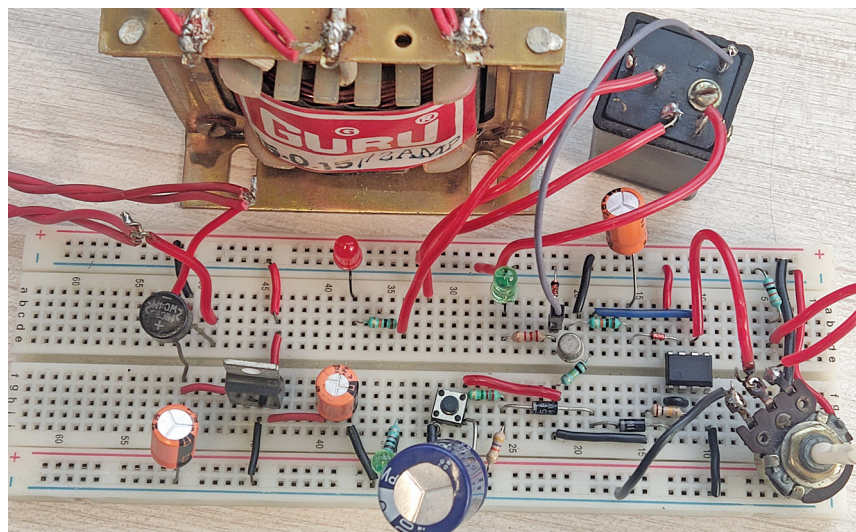


Fig. 1: EFY lab's prototype on a breadboard

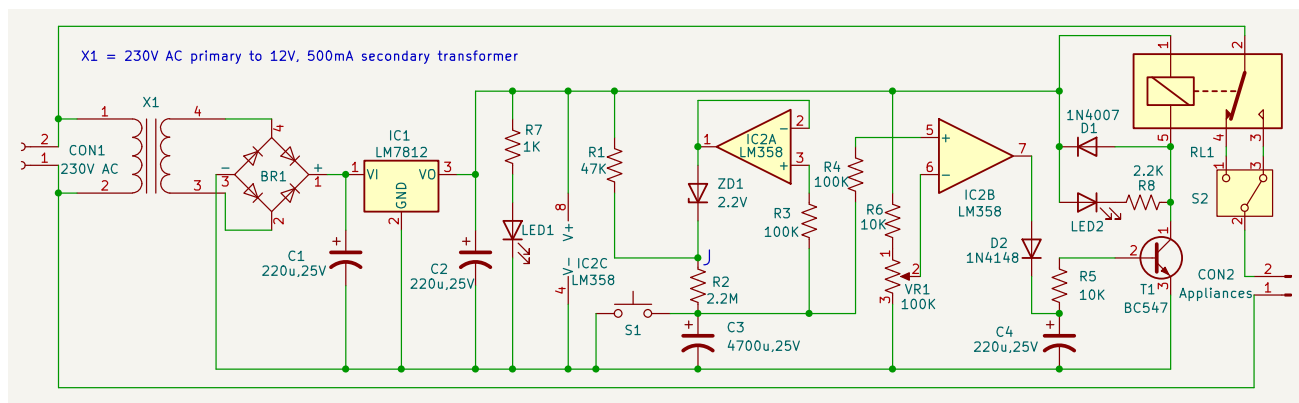


Fig. 2: Circuit diagram